SHUN ASO

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RESEARCH INTERESTS

Federated Learning · Large Language Models · AI for Mental Health · Human-AI Interaction · Ubiquitous Computing

EDUCATION

Keio University, Faculty of Environment and Information Studies (SFC) *Fujisawa, Japan*
B.A., Environment and Information Studies | Expected Graduation: July 2027

- Cumulative GPA: 3.7 / 4.0
- Research Lab: Nakazawa–Okoshi Laboratory (Ubiquitous Computing / HCI)
- Advisor: Prof. Tadashi Okoshi, Prof. Jin Nakazawa

International Baccalaureate Diploma Programme *Singapore*

- Score: 41 / 45 (Top 8% worldwide)
- 6 years of residence in Singapore; 5 years at an international school

RESEARCH EXPERIENCE

Undergraduate Researcher, Nakazawa–Okoshi Lab, Keio University SFC *2023 – Present*

- Research theme: Federated Learning (FL) applied to LLM fine-tuning for privacy-preserving mental health counseling systems
- Proposed Heterogeneity-Aware DualLoRA (HA-DualLoRA) and Selective Federated Aggregation (SFA), two novel FL fine-tuning strategies for realistic "Mixed IID/Non-IID" clinical federations, where general hospitals (IID) and specialized clinics (Non-IID) coexist
- Designed heterogeneity scoring using Shannon entropy and Otsu thresholding to classify clients without manual tuning
- SFA achieved the highest Non-IID counseling quality and personalization scores among all baselines while reducing per-client upload cost by 70%
- Independently conducted literature review, experimental design, implementation (Qwen2.5, Llama-3.2, LoRA/PEFT), and evaluation (7-metric LLM-judge framework) across the full research pipeline
- Open-sourced all code and experiment pipelines: github.com/shunasokeio/fllm-mentalhealth

Visiting Student, Anind Dey Lab, University of Washington iSchool *Aug 2026*

- Two-week research visit hosted by Prof. Anind K. Dey to explore collaboration on multimodal psychological state detection combining sensing and human-AI interaction signals
- Presented current federated learning research and discussed extensions toward Just-in-Time Adaptive Intervention (JITAI) frameworks

PUBLICATIONS & PRESENTATIONS

Shun Aso, Satoki Hamanaka, Jin Nakazawa, and Tadashi Okoshi. "Mental Health Counseling LLMs in Mixed IID/Non-IID Federations." Workshop Proceedings of the 2026 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp 2026), Shanghai, China, October 2026. (to appear)

Shun Aso, Satoki Hamanaka, Jin Nakazawa, and Tadashi Okoshi. "Fine-tuning LLM for Counseling Application in Federated Learning Setting." Adjunct Proceedings of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC 2025), Espoo, Finland, October 2025, pp. 100–104. (Poster)

Shun Aso, et al. "FLECS: Fine-tuning of LLM for Counseling Application Using Federated Learning." IPSJ Ubiquitous Computing Systems (UBI) Workshop, Osaka, Japan, February 2025. (Poster, in Japanese)

Additional Research Presentations

- Open Research Forum (ORF), Keio University — poster presentation
- Regional Well-being and DX Symposium — poster presentation
- "Smart City ABCs: Extending Behavioral Opportunity Beyond Bounded Rationality" Symposium — poster presentation
- Mental Health Workshop, UbiComp 2025 (Helsinki) — active discussant

GRANTS & AWARDS

- Yamagishi Student Project Support Grant, Keio University SFC (2025) — awarded for research proposal "Fine-tuning LLM for Counseling Applications Using Federated Learning"
- Best Paper Award Honorable Mention at "The 11th International Workshop onMental Health and Well-being", ACM UbiComp 2026, Shanghai

SKILLS & LANGUAGES

- Programming: Python, PyTorch, Hugging Face Transformers, Flower (FL framework), LoRA/PEFT, Git
- Languages: Japanese (native), English (fluent — TOEFL iBT 118/120)

OTHER EXPERIENCE

Private Tutor (Independent)*2023 – Present*

- Manages approximately 6 students across English, economics, mathematics, and science (IGCSE, A-Level, and Year 9–10 curricula)
- Designs original teaching materials and structured past-paper walkthroughs